CLASS 10 SCIENCE

HEREDITY

PART-1

* 1. INTRODUCTION TO HEREDITY

→ WHAT IS HEREDITY ?

Heredity is the process by which characteristics or traits are passed from parents to their offspring.

It ensures continuity of species by generating new individuals simil... to their parents.

→ WHAT ARE TRAITS ?

Traits are the observabl... of an organism.

Example – Eye ...

KEY POINT

...its make one organism similar to another.

→ VARIA...

Variation ... traits between individuals ...ecies.

For example ... persons have exactly ...ame height, face, fingerprint, etc.

→ TYPES OF VAR..ATION

(1) INHERITED VARIATION

- These variations are present in the genes.
- They are passed from parents to offspring.
- These variations exist from generation to generation.
- Example – Eye colour, blood group, height.

(2) ACQUIRED VARIATION

- These variations develop during an organism's lifetime.
- They are not present in the genes.
- They are not passed on to the next generation.
- Example – Scar on the body, tanned skin due to sun.

*** SUMMARY**

Heredity is the reason for the similarity in traits between parents and offspring, while variation creates differences among individuals. *

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HEREDITY

PART - 2

2. IMPORTANCE OF VARIATION

Variation is essential for the survival and progress of living organisms.

Variation is important for the following reasons :-

(1) Helps in Adaptation

- Variation helps organisms to adapt themselves to changes in the environment.
- Only those organisms that are better adapted can survive in the envir



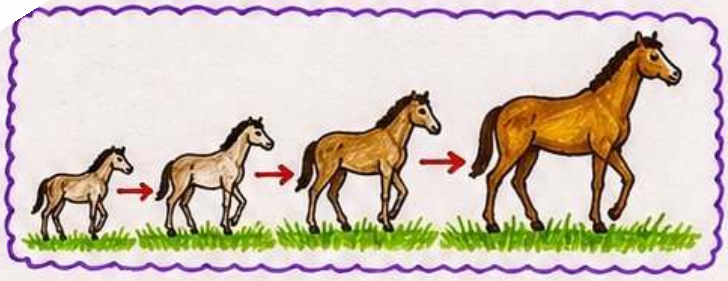
(2) Basis of Natural Sele

- Variation provides natural selectio
- Organisms survive elimina



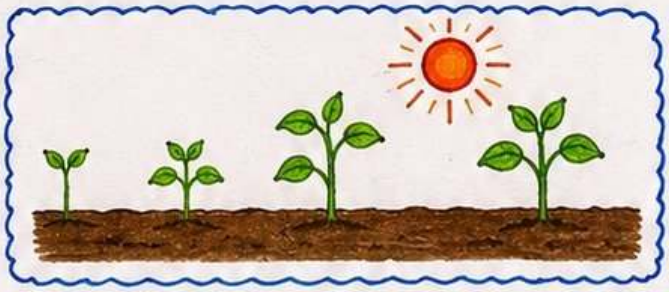
(3) Leads to

- Over a long ul variations acc
- This leads to t pment of new species. This ocess is called evolution.



(4) Ensures Survival of Species

- If there were no variation, all individuals would be alike.
- A sudden change in the environment might destroy all of them.
- Variation ensures that at least some individuals survive and the species continues.



* **EXAMPLE :** Some bacteria have variations which help them survive even when we take antibiotics. These bacteria are said to be antibiotic resistant. This variation is why the species continues to survive.

KEY POINT : → Variation is nature's way of ensuring that life continues in a changing world.



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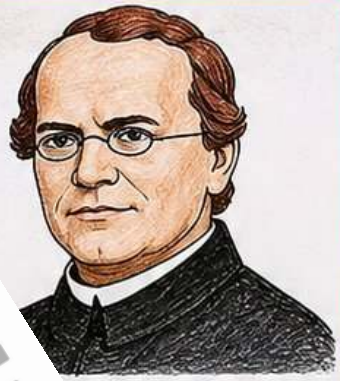
HEREDITY

PART - 3

3. MENDEL AND HIS CONTRIBUTION

Who was Mendel ?

- Gregor Johann Mendel (1822 – 1884) was an Austrian monk and scientist.
- He is known as the "Father of Genetics".
- He conducted experiments on pea plants and discovered the basic laws of inherita



Johann Mendel
? – 1884)

Mendel's Experiments

→ Mendel selected garden
for his experiments

→ He studied ho
are inher

→ He perform
observed the

LANT ?

grow.
ort life cycle.
* Many characters are clearly distinguishable.
* Normally self-pollinating (pure lines can be obtained).
* Produces a large number of seeds.

Mendel studied th ance of Seven Pairs of Contr ing Characters :

① Plant height : Tall (T) / Dwarf (t)
② Seed shape : Round (R) / Wrinkled (r)
③ Seed colour : Yellow (Y) / Green (y)
④ Flower colour : Purple (P) / White (p)
⑤ Pod shape : Inflated (I) / Constricted (i)
⑥ Pod colour : Green (G) / Yellow (g)
⑦ Flower position : Axial (A) / Terminal (a)

EXAMPLE OF SOME TRAITS

Tall	Dwarf
	
Round	**Wrinkled**
	

SUMMARY → Mendel's experiments on pea plants led to the discovery of the fundamental laws of inheritance which explain how traits are passed from parents to offspring.

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HEREDITY

PART - 4

4. MONOHYBRID CROSS

DEFINITION :

A cross which involves inheritance of only one pair of contrasting characters (one trait) is called a monohybrid cross.

EXAMPLE :

Let us consider the inheritance of plant height in pea plant.

Tall plant (TT) × Dwarf plant (tt)





(True-br… …eeding dwarf)

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✱ **P GENERATION (Parents) :**

Tall plant (TT) × Dwarf …

(from tt)

✱ **F_1 GENERATION :**

- Each offsp… each parent…
- All the F_1 of…
- This shows that … over dwarf (t).

…ants are Tall (Tt)

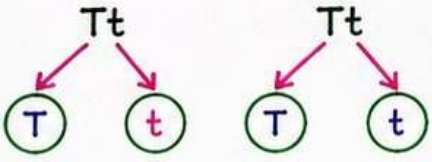


(Uniform / Same)

✱ **F_2 GENERATION (Se… …r F_1) :**

Cross : Tt × Tt

Gametes from each Tt plant :



Punnett Square :

(Female gametes) \ (Male gametes)	T	t
T	TT Tall	Tt Tall
t	Tt Tall	tt Dwarf

Result in F_2 Generation :

- Genotypic ratio : 1 TT : 2 Tt : 1 tt
- Phenotypic ratio : 3 Tall : 1 Dwarf (75% Tall : 25% Dwarf)

LAW OF DOMINANCE :

When two contrasting characters are brought together in a hybrid, one character (dominant) masks the expression of the other (recessive) character.

⇒

In this cross, Tall (T) is dominant and Dwarf (t) is recessive.

Hence, all F_1 plants are tall (Tt).

KEY POINT :

This cross shows the Law of Dominance and leads to the Law of Segregation.

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PART-5

5. LAW OF SEGREGATION

DEFINITION : The law of segregation states that the two alleles of a gene separate (segregate) during the formation of gametes and each gamete gets only one allele.

EXPLANATION :

- Every character is controlled by a pair of factors (alleles) – one from the father and one from the mother.
- These alleles remain together in the body (somatic cells) but get separated during the formation of gametes.
- So, each gamete receiv… one allele.
- During fertili… from male … restore the … offspring.

EXAMPLE (Using Tt plant) :

P GENERATI… (Parents)

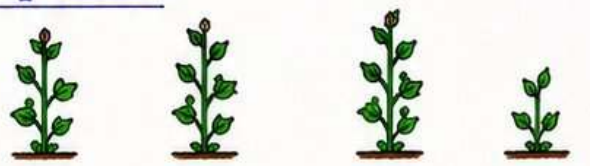
Tall pl… Tall plant (Tt)



T t

…GENERATION (Offspring)

F_2 Result :



Tall Tall Tall Dwarf

PUNNETT SQUARE

(Male gametes) → T t

(Female gametes)	T	t
T	TT Tall	Tt Tall
t	Tt Tall	tt Dwarf

→

1. GENOTYPIC RATIO : 1 TT : 2 Tt : 1 tt (1 : 2 : 1)

2. PHENOTYPIC RATIO : 3 Tall : 1 Dwarf (3 : 1)

IMPORTANT POINTS :

→ Alleles separate during gamete formation.
→ Each gamete gets only one allele.
→ Pair of alleles is restored at fertilisation.

KEY POINT :

* This law is also called Mendel's First Law of Inheritance.

SIGNIFICANCE OF LAW OF SEGREGATION :

- It explains why characters of parents reappear in the offspring.
- It is the basis of inheritance of traits from one generation to another.

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PART - 6

6. DIHYBRID CROSS

DEFINITION :

A cross which involves the inheritance of two pairs of contrasting characters (two traits) at a time is called a dihybrid cross.

EXAMPLE :

Let us consider the inheritance of seed shape and seed colour in pea plant.

Round Yellow (RRYY)	×	Wrinkled Green (rryy)
(True-breeding)		(True-breeding)

P GENERATION (Parents) :

Round Yellow (RRYY) × Wrinkled Green (rryy)

GAMETES FORMED :

RRYY → RY

rryy → ry

F_1 GENERATION (Offspring) :

- All the F_1 offspring are Round
- This shows that Round (R
  wrinkled (r) and Yel
  over Green (y).

F_2 GENERAT

Cross : RrYy

GAMETES FROM EA

RrYy → RY, Ry, rY, ry

Ry, rY, ry

Yellow (RrYy)

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	Ry	rY	ry
	RRYy Round Yellow	RrYY Round Yellow	RrYy Round Yellow
Ry	RRYy Round Yellow	RRyy Round Green	RrYy Round Yellow	Rryy Round Green
rY	RrYY Round Yellow	RrYy Round Yellow	rrYY Wrinkled Yellow	rrYy Wrinkled Yellow
ry	RrYy Round Yellow	Rryy Round Green	rrYy Wrinkled Yellow	rryy Wrinkled Green

RESULT IN F_2 GENERATION :

1. GENOTYPIC RATIO :

9 : 3 : 3 : 1

9 Round Yellow (R_ Y_)
3 Round Green (R_ yy)
3 Wrinkled Yellow (rr Y_)
1 Wrinkled Green (rr yy)

2. PHENOTYPIC RATIO :

9 : 3 : 3 : 1

9 Round Yellow
3 Round Green
3 Wrinkled Yellow
1 Wrinkled Green

IMPORTANT POINTS :

- In this cross, two traits (seed shape and seed colour) are inherited together.
- The genes for seed shape and seed colour are on different chromosomes.
- Hence, they assort independently.
- This gives four types of gametes = RY, Ry, rY, ry

LAW OF INDEPENDENT ASSORTMENT :

The law states that the alleles of one gene pair segregate (separate) independently of the alleles of another gene pair during the formation of gametes.

SIGNIFICANCE OF DIHYBRID CROSS :

→ It shows how two different characters are inherited independently.
→ It helps in understanding the basis of variation in living organisms.
→ It laid the foundation for the development of genetics.



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PART - 7

7. SEX DETERMINATION

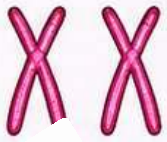
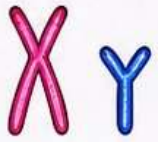
DEFINITION :

The process by which the sex (male or female) of an offspring is determined is called sex determination.

CHROMOSOMES IN HUMANS :

- Human body cells have 46 chromosomes.
- Out of these, 44 are autosomes and 2 are sex chromosomes.
- Sex chromosomes determine the sex of an individual.

SEX CHROMOSOMES :

FEMALE (XX)	MALE (XY)
	
44 + XX = 46	44 + XY = 46

MECHANISM OF SEX DETERMINATION IN HUMA

- Females (XX) are homogametic. They produc
  having X chromosome.
- Males (XY) are heterogametic. T
  one with X chromosome and
- At the time of fertilis
  X bearing sperm
- This decides w

y (XY).

METES FORMATION

FEMALE (XX)	MALE (XY)
	Testes
X	X Y
All eggs carry X	Sperms carry X or Y

PUNNETT SQUARE

Father (XY) / Mother (XX)	Egg: X	Egg: Y
X	XX (Girl)	XY (Boy)
X	XX (Girl)	XY (Boy)

RESULT :

- If sperm with X chromosome fertilises the egg (X) → XX (Girl)
- If sperm with Y chromosome fertilises the egg (X) → XY (Boy)
- Chances of a girl = Chances of a boy = 1 : 1



Girl (XX) | Boy (XY)

IMPORTANT POINTS :

- The mother always contributes an X chromosome.
- The father contributes either X or Y chromosome.
- The sex of the baby depends on the sperm (father).
- The ratio of boys to girls is 1 : 1.

SUMMARY :

- Females are homogametic (XX) → only X gametes.
- Males are heterogametic (XY) → X and Y gametes.
- XX combination gives a girl.
- XY combination gives a boy.
- Thus, the father determines the sex of the child.

KEY POINT : It is the sperm that decides whether the baby will be a BOY or a GIRL.

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CLASS 10 SCIENCE
HEREDITY

PART-8
SUMMARY & KEY POINTS

8. COMPLETE CHAPTER SUMMARY

1. HEREDITY
→ Heredity is the process by which traits are passed from parents to offspring.
→ It ensures continuity of species and creates similarities among family members.

2. VARIATION
→ Variation means differences between individuals.
→ Types of Variation :
- Inherited Variation – passed from parents to offspring.
- Acquired Variation – developed during life due to environment and not inherited.

3. MENDEL AND HIS CONTRIBUTION
→ Gregor Johann Mendel is known as the "Father of Genetics".
→ He conducted experiments on pea plants.
→ Mendel selected pea plant because :
- Short life cycle
- Easy to grow
- Many contrasting characters
- Self-pollinating (pure lines can be obtained)

→ He studied 7 pairs of contrasting traits in pea plant.

4. MENDEL'S LAWS OF INHERITANCE
① Law of Dominance : In a hybrid, one trait (dominant) masks the expression of the other (recessive) trait.
② Law of Segr… two alleles of a gene separate …on of gametes and each …llele.
③ …nt : The alleles …e) independently during the

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(A) MONOHYBRID CROSS
Example : Tall (TT) × …
P Generation : …
Gametes :
F_1 Generation : A…
F_2 Generation : Tt
Result :

	T	t
T	TT	Tt
t	Tt	tt

… :
1 Dwarf
…3 : 1)

(Two traits)
…(RRYY) × Wrinkled Green (rryy)
… Round Yellow (RrYy)
…: RrYy × RrYy
…rom each F_1 plant : RY, Ry, rY, ry
…ut in F_2 Generation :
Genotypic ratio : 1 : 2 : 1 : 2 : 4 : 2 : 1 : 2 : 1 (9 types)
Phenotypic ratio : 9 : 3 : 3 : 1
9 Round Yellow : 3 Round Green :
3 Wrinkled Yellow : 1 Wrinkled Green

6. SEX DETERMINATION IN HUMAN BEINGS
→ Human body cells have 46 chromosomes.
→ Out of these, 44 are autosomes and 2 are sex chromosomes.
→ Females (XX) are homogametic – produce only X type of gamete.
→ Males (XY) are heterogametic – produce two types of gametes, X and Y.
→ At fertilisation :
X (from mother) + X (from father) = Girl (XX)
X (from mother) + Y (from father) = Boy (XY)

Egg (Mother) \ Sp… (Father)	X	Y
X	XX (Girl)	XY (Boy)
X	XX (Girl)	XY (Boy)

Ratio of Girls : Boys = 1 : 1

7. KEY POINTS
* Traits are controlled by genes.
* Different forms of a gene are called alleles.
* Dominant trait is expressed in F_1 generation.
* Recessive trait reappears in F_2 generation.
* During gamete formation, alleles segregate.
* Genes for different traits assort independently.
* Sex of the child is determined by the father's sperm.
* Mendel's work laid the foundation of genetics.

8. SUMMARY IN ONE LINE
① Heredity passes traits from parents to offspring.
② Variation is essential for survival and evolution.
③ Mendel discovered the laws of inheritance.
④ Traits are controlled by genes and alleles.
⑤ Alleles segregate during gamete formation.
⑥ Genes for different traits assort independently.
⑦ Father determines the sex of the child.

IMPORTANT TERMS
Trait – A characteristic feature | Gene – Unit of heredity | Allele – Different form of a gene | Dominant – Expressed in hybrid
Recessive – Masked in hybrid | Gamete – Sex cell (sperm or egg) | Genotype – Genetic constitution | Phenotype – Physical appearance
Autosome – Non-sex chromosome | Sex chromosome – Determines sex